

Day 7 — When To Use It

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify problems where nikhilam is faster than long multiplication.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: ādhikya; lit. “excess (number above base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The decision tree.**
 - Both factors within ~5 of the same base → Nikhilam is fast.
 - One factor close to base, other far → traditional multiplication often faster.
 - Both factors far from any base (e.g. 47×63) → traditional.
 - Both factors near DIFFERENT bases (e.g. 98×1005) → tricky; pick a common base and scale.
2. **Speed comparison.** Time 97×96 vs 47×63 mentally. Most students will find Nikhilam dominates the first; long multiplication dominates the second.
3. **Pedagogical limit.** Nikhilam multiplication is NOT a universal speed-up. It’s a tool for ONE problem class. Recognising when to use it is the skill.
4. **Practice with strategic choice.** 10 problems, mixed. Each student picks the best method and notes it next to the answer.
5. **Reflection.** “*What pattern of problems makes Nikhilam shine? What pattern defeats it?*” Each student writes 3 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces When To Use It.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).